

Vadodara Institute of Engineering

MIA ATL Report for Phase -1 (9-1-2026 to 8-4-2026)

Name of School	Navrachana Vidhyani Vidhyalaya
Principal of School	Dr. Archana
Nodal Teacher of School	Jinny Shah
RMOC	Mr. Mukesh Bind
VIER Mentor Faculty	Naman Shah
Core Team Leader	Jayraj Chavda
VIER Interns	(1) Henil Amin (2) Shanvi Patel (3) Naiya Patel
Total ATL visit days	(1) 11 (2) 10 (3) 1
Total Atal sessions	(1) 11 (2) 10 (3) 1

(A) Projects

Titles & Objectives	Solar Setu Project - helps students understand the working of different sensors like light, temperature, and voltage sensors, and how they collect environmental and solar energy data for practical applications. - It teaches real-time data reflection by using platforms like NodeMCU ESP8266, where sensor data is sent to cloud dashboards for live monitoring and analysis.
Use of Hardware/ Components / Tools	Solar Panel Sensor measures sunlight intensity, voltage, or current generated by the panel to monitor solar energy production and system efficiency. - Arduino Uno processes sensor data and controls the system operations in the project. - NodeMCU ESP8266 enables wireless communication, sending real-time solar data to online platforms for monitoring and analysis.
Use of Software	- Arduino IDE is used to write, compile, and upload code to microcontrollers, helping students control sensors and manage real-time data in IoT projects like Solar Setu. - Code.org helps students learn basic programming concepts through interactive lessons, building a strong foundation for developing IoT-based applications.
Suggestions / Improvement / Solution provided	Implement real time data reflection for the amount of solar energy generated Data Collection & Processing Use a solar panel sensor connected to Arduino Uno to measure voltage/current. The Arduino processes this data and calculates power (energy output).

	2. Wireless Data Transmission & Display Send the processed data to NodeMCU ESP8266, which uploads it to cloud platforms (like dashboards or apps). This allows students to view real-time solar energy generation on their mobile or computer.
(B) Innovative topics / technologies discussed: Machine learning, Android, Web Development	

Name & Sign of VIER Mentor Faculty with date: Naman Shah 20/04/2026